Macromolecules
Cite This:
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Comprehensive Structural Assessment of Linear Block Polymers by NMR and SEC

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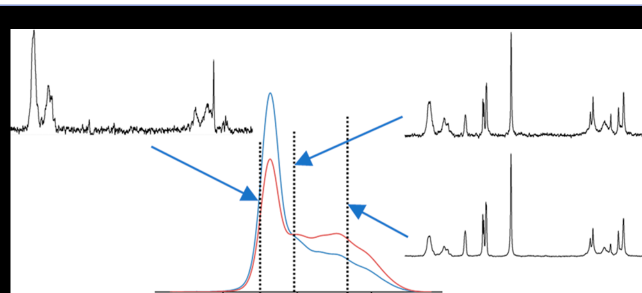
ABSTRACT:



1. INTRODUCTION

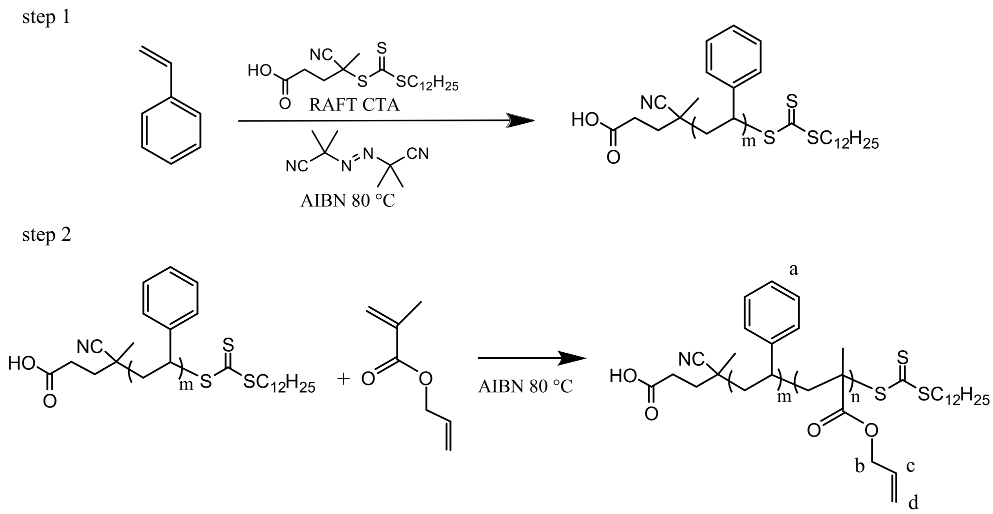
Block polymers are used for a broad range of applications. Their tailored structure leads to unique functions not provided by homopolymers or random copolymers. Linear diblo polymers, for e ple, have found e in dispersion, ' compatibilization, ' and drug delivery. ' Controlled radical polymerization (CRP) techniques such as nitroxide-mediated polymerization (NMP , reversible addition−fragmentation chain transfer (RAFT), and atom transfer radical polymerization (ATRP) are commonly used today because of their versatility and ease of application. Synthetic methods can involve a two-step process where one block is formed before the other is added onto it. Other methods involve a one-step process that favors the formation of one block early in the reaction, followed by a second block formation later. The simplicity of a one-step synthesis is advantageous but relies upon a significant difference in reaction rates of individual pairs if a tailored polymer is to form. For example, polymers synthesized with styrene (S) and maleic anhydride (MA) show a dramatic increase in the rate of polymerization with increasing MA feed content. Because MA does not self-polymerize under normal conditions, chains rich in S-*alt*-MA form early in the polymerization process, followed by S-rich chains. Common problems can occur in either synthetic approach such as homopolymer formation and compositional variation (often as a function of molecular weight). These complications lead to heterogeneous materials that may fail in function or otherwise lead to unwanted effects. It is important to establish an analytical strategy that is efficient and effective in providing a detailed characterization of the material to guide synthesis and purification.

Size exclusion chromatography (SEC) and nuclear magnetic resonance (NMR) are routine methods that provide the detailed information necessary for a thorough materials characterization. SEC provides molecular weight (M) distribution and estimates for weight-average molecular weight (M_w), number-average molecular weight (M_n), and polydispersity index (PDI). This is helpful to show molecular weight growth but does not provide chemical structure information. However, coupled with a multidetection scheme, in some cases, SEC can provide clues to composition drift as a function of molecular weight. ^{1}H NMR is generally used to obtain polymer chemical composition, but in many cases involving polymeric materials, the spectra are too broad to obtain accurate results. Quantitative ^{13}C NMR can be more useful because of the better chemical shift dispersion. NMR may be used to determine M_n for one or more blocks if an end group or connecting group is spectrally resolved. Although routine NMR can provide overall detailed chemical structure composition and sequencing information, it cannot be used to strictly identify heterogeneous mixtures or distinguish compositional variation with molecular weight. ^{1}H pulsed gradient spin-echo (PGSE) NMR provides the ability to obtain chemical information based upon the material's diffusion coefficient (. he technique not only can be used to estimate M_w, − but with the appropriate data processing method it can be used to identify compositional variation molecular weight including homopolymer formation. '

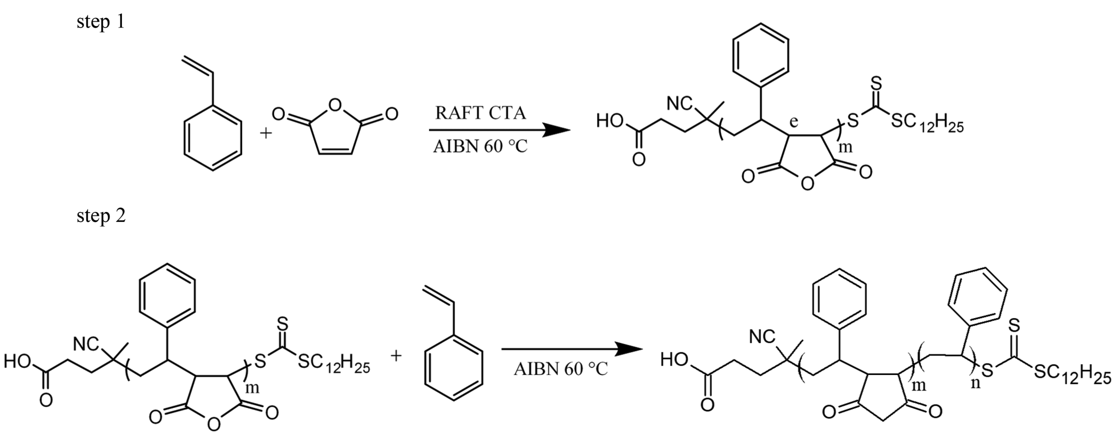
Received: October 22, 2018
Revised: December 13, 2018
Published: January 25, 2019

DOI:
Macromolecules 2019, 52, 1025−1032

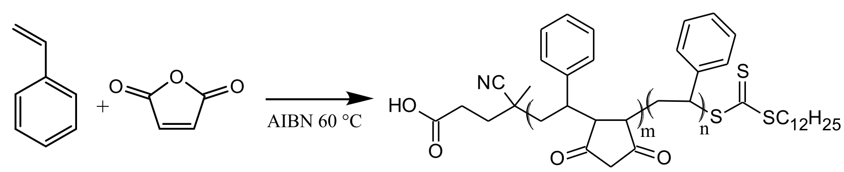
Scheme 1. Two-Step Synthesis of Poly(styrene-*b*-allyl methacrylate) via RAFT Polymerization



Scheme 2. Two-Step Synthesis of Poly(styrene-*b*-(styrene-*alt*-maleic anhydride)) via RAFT Polymerization



Scheme 3. One-Step Synthesis of Poly(styrene-*b*-(styrene-*alt*-maleic anhydride)) via RAFT Polymerization



Several univariate and multivariate data processing methods are available. In our lab, we favor the use of direct exponential curve resolution algorithm (DECRA). − This processing method resolves component spectra along with their respective diffusion coefficients and is therefore useful in assessing chemical composition as it varies with molecular weight. Especially useful is its ability to quantitatively resolve homopolymer from copolymer despite complete spectral overlap. This ability to analyze materials with minimal cleanup steps helps to expedite the synthesis campaign and guide further purification.

We present the detailed characterization of three polymers: poly(styrene-*b*-allyl methacrylate) (S-AM) synthesized by a two-step RAFT method and poly[styrene-*b*-(styrene-*alt*-maleic anhydride)] (S-S/MA) synthesized by both two-step and one-step RAFT methods. By combining all three techniques (SEC, 1D NMR, and ^{1}H PGSE NMR + DECRA), we establish a comprehensive characterization of linear block polymers.

2. EXPERIMENTAL SECTION

2.1. Materials. Styrene (S, 99%), maleic anhydride (MA, 99%), allyl methacrylate (AM, 98%), RAFT chain transfer agent 4-cyano-4-[(dodecylsulfanylthiocarbonyl)sulfanyl]pentanoic acid (CTA, 97%), and initiator azobis(isobutyronitrile) (AIBN, 98%) were obtained from Sigma-Aldrich. S and AM were purified using inhibitor removal columns from Sp^2 to remove *tert*-butyl catechol and hydroquinone monomethyl ether, respectively. MA and AIBN were purified by recrystallization from chloroform and methanol, respectively. CTA was used as received. The solvent, tetrahydrofuran (THF), was obtained from Sigma-Aldrich and used as received. The solvents 1,4-dioxane, diethyl ether, dimethylformamide (DMF), petroleum ether, and methanol were obtained from VWR and used as received. The deuterated NMR solvents CD_2Cl_2, acetone-d_6, and THF-d_8 were obtained from Cambridge Isotopes Laboratory and used as received.

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2.2. Polymer Synthesis. Three polymers were synthesized for analysis. The block copolymer, poly(styrene-*b*-allyl methacrylate) (S-AM), was synthesized by a two-step RAFT method. Two block polymers, poly(styrene-*b*-styrene-*alt*-maleic anhydride) (S-S/MA), were synthesized by both two-step and one-step RAFT methods. All representations of composition are as molar ratios and not degrees of polymerization.

2.2.1. S-AM (Two-Step Method; Aim S_{43}-b-AM_{57} Molar Ratio). Step 1: Styrene (33.4 g), CTA (96.4/1 molar ratio [S monomer]/CTA), and AIBN (1830/1 molar ratio [S monomer]/AIBN) were added into a 100 mL single-neck round-bottom flask. The mixture was purged by a freeze−thaw method to remove oxygen. The flask was placed in a bath of dry ice/acetone until the contents were frozen. Using a Firestone valve, the flask was evacuated using a house vacuum for a few minutes. The valve was then switched to introduce nitrogen, and the contents thawed using a bath of water. The process was repeated. The flask was then placed in a preheated oil bath at 80 °C overnight. The flask was then cooled, and contents were precipitated into methanol. The precipitate was then filtered and dried overnight in a high-vacuum oven at room temperature. The yellow solid was redissolved in THF at 30% solids, precipitated into methanol, collected by filtration, rinsed with methanol, and dried in a high-vacuum oven at room temperature overnight.

Step 2: Polystyrene (3.00 g from step 1 with attached CTA), allyl methacrylate (4.79 g; 1.32/1 molar ratio [AM monomer]/[S monomer]), and AIBN (832/1 molar ratio [AM monomer]/AIBN) were added to THF (30% solids) into a 100 mL single-neck round-bottom flask. The mixture was capped with a septum and purged with nitrogen gas for 60 min to remove oxygen before being placed in a preheated oil bath at 67 °C overnight. The solution was cooled, precipitated into methanol, filtered, and dried. The solid was then dissolved in THF overnight, precipitated into methanol, filtered, and dried a second time.

2.2.2. S-S/MA (Two-Step Method; Aim S_{20}-b-S_{40}/MA_{40} Molar Ratio). Step 1: Styrene (2.00 g), maleic anhydride (1.88 g; 1/1 molar ratio [AM monomer]/[S monomer]), CTA (148/1 molar ratio [S monomer]/CTA), and AIBN (3000/1 molar ratio [S monomer]/AIBN) were added to dioxane (48% solids) into a 100 mL single-neck round-bottom flask. The mixture was capped with a septum and purged with nitrogen gas for 45 min in an ice bath to remove oxygen before being placed in a preheated oil bath at 60 °C for ∼69 h. The viscous solution was cooled and precipitated into diethyl ether. The solid was dried in a high-vacuum oven at room temperature for a few hours.

Step 2: Poly(styrene-*alt*-maleic anhydride) (3.60 g from step 1 with attached CTA) was added to DMF (39% solids) into a 100 mL single-neck round-bottom flask until dissolved. Styrene (0.930 g; 1/2 molar ratio [added S monomer]/[S-MA dimer]), and AIBN (148/1 molar ratio [added S monomer]/AIBN) were then added (45% solids total). The solution was placed on ice and purged with nitrogen gas for 45 min to remove oxygen before being placed in a preheated oil bath at 60 °C for 66 h. The viscous solution was cooled, diluted with acetone, precipitated into petroleum ether, and dried in a high-vacuum oven for about 2 h. The material was redissolved in acetone, precipitated in petroleum ether, filtered, and then dried in the high-vacuum oven at room temperature overnight.

2.2.3. S-S/MA (One-Step Method; Aim S_{58}-b-S_{21}/MA_{21} Molar Ratio). Styrene (1.20 g), maleic anhydride (0.300 g; 1/4 molar ratio [MA monomer]/[S monomer]), CTA (384/1 molar ratio [S monomer]/CTA), and AIBN (860/1 molar ratio [S monomer]/AIBN) were added to dioxane (50% solids) into a 50 mL single-neck flask. The solution was capped with a septum and purged with nitrogen gas for 45 min in an ice bath to remove oxygen before being placed in a preheated oil bath at 60 °C for ∼65 h. The viscous solution was cooled, diluted with acetone, precipitated into petroleum ether, filtered, and dried in a high-vacuum oven at room temperature for a few hours. The material was redissolved in acetone, precipitated in petroleum ether, filtered, and then dried in the high-vacuum oven at room temperature for a few hours.

2.3. SEC. The column set used includes three 7.5 × 300 mm^2 Plgel mixed-C (for S-AM) or mixed-B (for S-S/MA) columns from Polymer Laboratories (Agilent). Detectors were arranged in the following order: Waters 2487 UV−vis detector (270 nm), Varian (Agilent) 390-LC two-angle light scattering detector (for S-AM), or Precision Detectors PD2000 15° and 90° light-scattering (LS) detector (for S-S/MA), parallel split between a Viscotek H502A differential viscometer (DV) and a Waters 2414 differential refractive index (DRI) detector. The columns, LS, DV, and DRI detector temperatures were 35.0 ± 0.1 °C. The eluent was HPLC-grade uninhibited THF for S-AM and THF + 1% formic acid (THF/FA) for the S-S/MA polymers, delivered at a nominal flow rate of 1.0 mL min^{-1}. Flow rate corrections were made from the retention volume of acetone added to the sample solvent at a concentration of 0.2% (v/v). A universal calibration curve was constructed using 16 narrow-molecular-weight distribution polystyrene standards (Polymer Laboratories/Agilent) ranging from a molecular weight of 162 to 1920000. Samples were injected in a volume of 100 μL at a concentration of 2.0 mg mL^{-1} for molecular weight measurement by viscometry and light-scattering detection. Results are plotted as the normalized differential logarithmic molecular weight distribution where the ordinate, W_n(log M), is the weight fraction of polymer per log M increment.

2.4. ^{1}H PGSE NMR. Experiments were performed at 23 °C on a Varian/Agilent Inova spectrometer operating at a ^{1}H frequency of 500 MHz and having a probe capable of producing constant gradient fields up to 60 G/cm. S-AM samples were dissolved at ∼0.1 wt % in both CD_2Cl_2 and THF-d_8, and S-S/MA samples were dissolved at ∼0.2 wt % in both acetone-d_6 and THF-d_8/FA. The pulse sequence employed is the bipolar stimulated-echo pulse sequence with thermal convection current compensation. Typically, 10 gradient values (g) ranging from 5 to 55 G/cm were used with 64−256 transients for signal averaging, resulting in an experimental time between 1 and 3.5 h. DECRA analysis requires that g be varied in the experiment such that the increment in g^2 is a constant. Furthermore, the number of components that is resolved in the DECRA analysis is a user input and not determined automatically.

Using PGSE NMR to measure the solution size of a component with a discrete M is straightforward, but there are limitations in measuring components with size distributions, such as polymers. Because the technique uses no physical separation, obtaining the true M distribution curve, one that is multimodal or exhibits other complex structure, is not possible under most circumstances. The technique therefore provides an approximation to the true distribution found by SEC. Several data processing approaches are available that can approximate M distributions may be used. Most common are methods that derive the M distribution from the inversion of the Laplace transform (ILT) using CONTIN or maximum entropy. A 2D plot with diffusion coefficient (or M_w) on the ordinate and the 1D NMR spectrum on the abscissa is established that, in principle, can show compositional variation with M_w. But, this approach can be vulnerable to numerical instability. A large number of spectra covering a full range of signal attenuation must be acquired, and the experiments can be time consuming.

Other techniques, such as DECRA and SCORE (Speedy Component Resolution), provide discrete spectra with associated M_w values. If the polydispersity is large enough, several components may be resolved, typically 2−3, which accommodate the molecular weight range. Each spectrum is representative of the material found in that M region. This feature of the analysis is particularly useful in determining compositional drift in multicomponent polymers and identifying homopolymer formation. DECRA and SCORE perform with similar characteristics. Much fewer spectra are typically required relative to the ILT methods, and the processing time is much shorter. Both require user input of the number of resolved components, but there are important differences to note. Unlike DECRA, SCORE has no sampling restrictions and can accommodate decay functions different from pure exponentials, such as probe-dependent corrections for gradient nonuniformity. The effect of nonexponential and multiexponential decay behavior on DECRA analysis has been investigated in detail.

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Although PGSE NMR is not strictly quantitative, it is usually valid to compare integrals of specific chemical groups among the resolved components. For example, it is likely that the relaxation behavior of an aromatic CH group in a low-M PS chain is very similar to that found in a high-M PS chain. The integral of this group may then be compared among resolved spectra. However, a backbone CH_2 group may have very different relaxation behavior compared with an aromatic CH and therefore may not be accurately compared within the same resolved spectrum.

2.5. ^{13}C NMR. ^{13}C NMR spectra of the S-S/MA polymers were obtained at 23 °C on a Varian/Agilent Inova spectrometer operating at a ^{13}C frequency of 125 MHz using a standard 10 mm probe. Samples were dissolved in acetone-d_6 at 40 mg mL^{-1} with Cr(acac)$_3$ and tetramethylsilane (TMS) added as a relaxation agent and chemical shift reference, respectively. Gated decoupling with a 3 s recycle delay were used to obtain quantitative spectra. A delay of 3 s was determined to be sufficient for quantitative assessment (, Figure S2). Transients numbering 10240 were accumulated over a total experimental time of ~9 h.

2.6. Polymer Solution Behavior. Important considerations are necessary in determining molecular weight of polymers via PGSE NMR. The solution hydrodynamic radius (R_h) of a polymer chain is affected by solvent quality. In a good solvent the polymer chains and solvent molecules interact favorably and swell under excluded volume effects. In a poor solvent, the polymer chain will collapse as the chains prefer self-interactions over that with the solvent. In the middle is the θ condition, where there are no excluded volume effects and the polymer conformation follows the ideal chain model. Still other conditions may affect chain size in solution. For example, Coulombic charges along the backbone may cause swelling unless they are screened by dissolved ions. Block polymers may exhibit the complication that one block interacts significantly different with the solvent than does the other block.

To estimate M_w from D, we must first account for the effects of solvent viscosity (η) and temperature (T). If we assume the dissolved polymer chain under dilute conditions behaves as a spherical particle in a fluid continuum, we may use the Stokes–Einstein equation to estimate size as follows:

$$D = kT/6\pi\eta R_h \tag{1}$$

where k is the Boltzmann constant. The temperature-dependent solvent viscosity is calculated from empirical formulas based on protonated solvents. The relationship between size and molecular weight has been shown to follow a scaling relationship:

$$R_h \propto M^{\nu} \tag{2}$$

The scaling parameter has been related to solvent quality where $\nu \sim 0.6$ for a polymer in a good solvent and ~0.5 for that in a θ solvent. A practical list of experimental values of ν for several polymer–solvent combinations is found here. M_w is obtained by first determining R_h from then using the scaling behavior from . We use the scaling of poly(ethylene oxide) (PEO) in D_2O at 25 °C with an exponent $\nu = 0.55$. The rational is based on efficiency. Given the broad range of polymer types and limited economic deuterated solvents available, this choice affords a decent approximation for most studies in our laboratory. The final empirical relation to convert R_h to M_w is $M_w = 3.122 \times 10^{19}(R_h \times 10^{-9})^{1.832}$, where the unit of R_h is m. It is imperative to be in the dilute regime where there are negligible nearest-neighbor interactions. For a polymer less than $M_w \sim 50000$, a concentration of 0.5% w/w is acceptable. Higher molecular weight requires further dilution; between $M_w \sim 50000$ and 1000000 a concentration of 0.05% w/w is acceptable.

3. RESULTS AND DISCUSSION

Materials from both steps of the S-AM synthesis are compared in the SEC overlay in . The copolymer curves exhibit a bimodal character in which the low molecular weight mode lines up with the material from the first step in the reaction. S homopolymer likely exists, indicating that a fraction of the RAFT end groups was unreactive. The presence of the S homopolymer is confirmed by the DECRA analysis, shown in . Resolved component 1 shows only S resonances and has similar M_w as the material from step 1 of the reaction. A summary of M_w values measured by SEC and calculated from the DECRA analyses of PGSE NMR data is compiled in , including data from several solvents. A simple integration of the styrene aromatic resonance (6–7.5 ppm) in gives a valid estimate of the amount of unreacted homopolymer. Although PGSE NMR is not strictly quantitative, we can make the exception when analyzing a single resonance envelope among the resolved spectra. We can assume in this case that the molecular motion of the styrene ring is similar among all polymer chains and therefore exhibits similar relaxation behavior. The full molar composition as determined from 1D ^1H NMR (, Figure S1) is found to be S/AM = 55/45. However, from the DECRA analysis we see that 55% of the S integral is from homopolymer. The more accurate representation of the final material is $S_{31} + S_{25}$-b-AM_{44} (molar ratio). The final S-AM copolymer, therefore, has a molar composition of S/AM = 36/64. The resolved M_w values are shown in as vertical lines with heights proportional to component concentrations and reasonably span the SEC curves. The analyses in two solvents, THF-d_8 and CD_2Cl_2, show similar results ( ). Because the NMR analysis derives M_w from the same polymer calibration, PEO in D_2O, the solvent quality for these two solvents must be similar. The NMR data closely aligns with SEC but is not exact. Because the data from both SEC and NMR were obtained in the same solvent, THF, this is indicative of a mismatch in the polymer calibration used for both techniques.

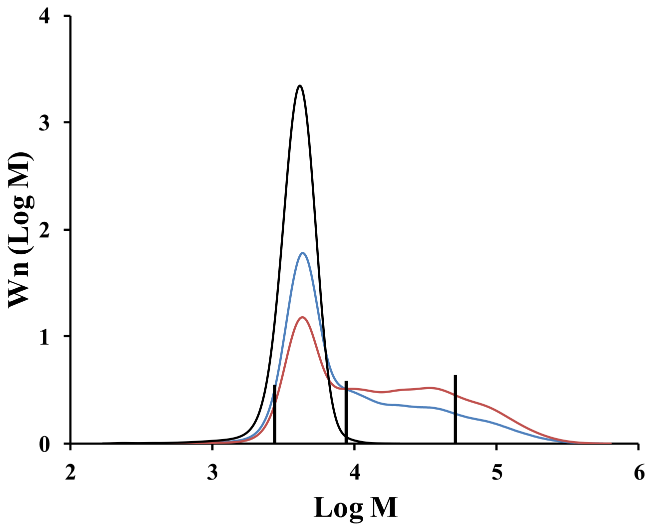

Figure 1. SEC curves of S-AM in THF. S prepolymer is shown in black. S-AM block copolymer is shown with UV detection (270 nm) in blue and DRI detection in red. The curves are normalized such that each represents equally the same total polymer weight. Resolved, discrete M_w values from the DECRA analysis of the step 2 material in THF-d_8 are shown as vertical lines with heights proportional to component concentrations (from).

Compositional drift with molecular weight in the copolymer is evidenced in both techniques. Because of the high extinction coefficient and lower wavelength position of λ-max, the UV signal from S is emphasized over the other components at 270 nm detection in SEC (see). The difference found between UV and DRI detection indicates a loss in S relative to

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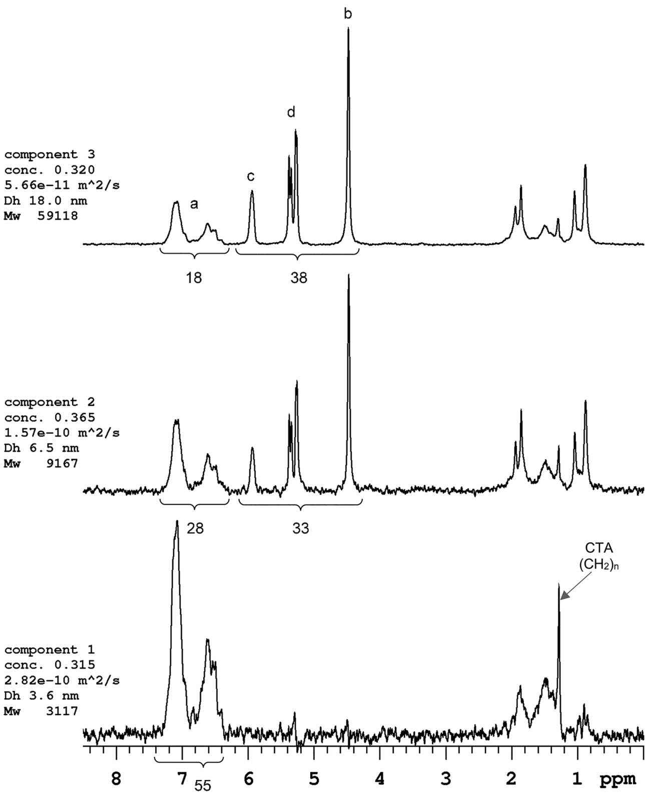

Figure 2. DECRA analysis of S-AM in CD_2Cl_2. Three polymeric components are resolved and scaled individually. Component 1 is S homopolymer. Components 2 and 3 are discrete S-AM components that are representative of the broad molecular weight distribution. Integral estimates are shown below each component spectra. The concentration (conc) represents the total spectral integral contribution of that component to the original data set. The hydrodynamic diameter (D_h, nm) is calculated from the measured diffusion coefficient ($m^2\ s^{-1}$, given above the D_h value), the solvent viscosity, and the experimental temperature using the Stokes−Einstein equation. The M_w values are estimated from the M_w vs D_h relationship of PEO in D_2O at 25 °C. The solvent resonances are ignored in the analysis.

AM as molecular weight increases (seen between log $M \sim 3.9$–4.7). A similar conclusion can be drawn from the DECRA analysis. Component 3 (higher M_w) shows a higher AM content relative to component 2. The integral ratio, S/AM, is ~45% lower in component 3 relative to component 2.

Materials from both steps of the two-step S-S/MA synthesis are compared in the SEC data in . The obvious failure

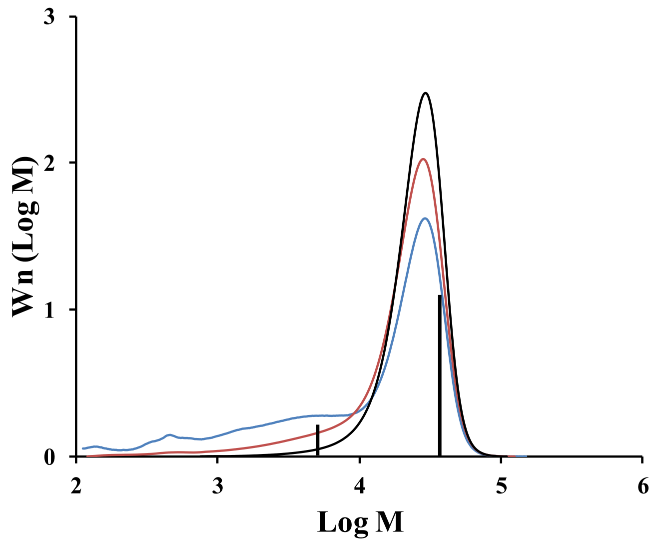

Figure 3. SEC curves of S-S/MA (two-step method) in THF/FA. S/MA prepolymer is shown in black. S-S/MA block copolymer is shown with UV detection (270 nm) in blue and DRI detection in red. The curves are normalized such that each represents equally the same total polymer weight. Resolved, discrete M_w values from the DECRA analysis of the step 2 material in THF-d_8/FA are shown as vertical lines with heights proportional to component concentrations (from).

to increase molecular weight in the reaction is indicative of a failed synthesis, but further materials characterization may be obtained. The differences between the two materials are better highlighted in the UV detection curve that exhibits an enhanced signal at the lower molecular weight region (log M < 4). Again, the difference between UV detection and DRI detection is indicative of higher S content. Because only S was added in the second step, we deduce from the SEC that low molecular weight S homopolymer formed. The structure of the low molecular weight material is confirmed by ^{1}H PGSE NMR. DECRA results in reveal a resolved low molecular weight component composed primarily of S. The corresponding calculated M_w values of the two components line up well with the SEC in . These results clearly indicate a failure to create the S block.

Table 1. Molecular Weight Data Obtained by SEC and NMR for the Polymers in Various Solvents

		SEC		PGSE NMR with DECRA analysis			
polymer		M_w	solvent	M_w (component concentration)			solvent
S-AM	step 1	4100	THF	3200			THF-d_8
				2800			CD_2Cl_2
	step 2	31000	THF	2800 (0.31)	8700 (0.33)	52000 (0.36)	THF-d_8
				3100 (0.32)	9200 (0.37)	59000 (0.32)	CD_2Cl_2
S-S/MA	step 1	27000	THF/FA	31000			THF-d_8/FA
				26000			acetone-d_6
	step 2	24000	THF/FA	5000 (0.16)	38000 (0.84)		THF-d_8/FA
				4200 (0.17)	33000 (0.83)		acetone-d_6
S-S/MA	step 1	52000	THF/FA	23000 (0.37)	66000 (0.63)		THF-d_8/FA
				19000 (0.48)	53000 (0.52)		acetone-d_6

[a]Component concentration represents the total spectral integral contributed by that component to the overall integral of the mixture. Data not presented in the main article are included in the .

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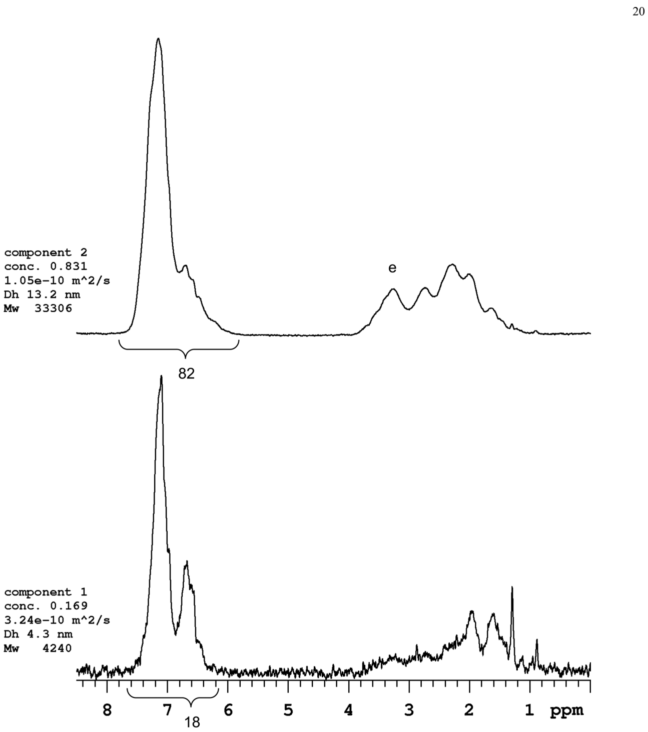

Figure 4. DECRA analysis of S-S/MA (two-step method) in acetone-d_6. Two components are resolved and scaled individually. Aromatic integral estimates are shown below each component spectra. Component 1 is primarily the low molecular weight S homopolymer where very little MA is observed. Component 2 is primarily the S/MA material from Step 1. The solvent resonances are ignored in the analysis.

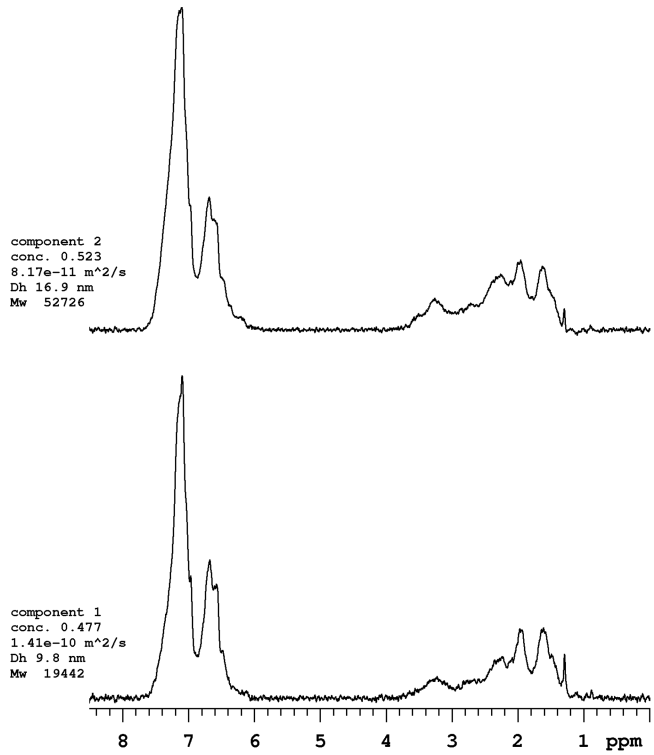

Figure 6. DECRA analysis of S-S/MA (one-step method) in acetone-d_6. Two components with very similar spectra are resolved and scaled individually. Slightly more MA signal (2.2−3.8 ppm) relative to S is seen in component 2, indicating slight compositional drift. The solvent resonances are ignored in the analysis.

The results for the one-step approach are shown in and . The SEC overlay of the two detection methods shows no evidence of compositional heterogeneity with molecular

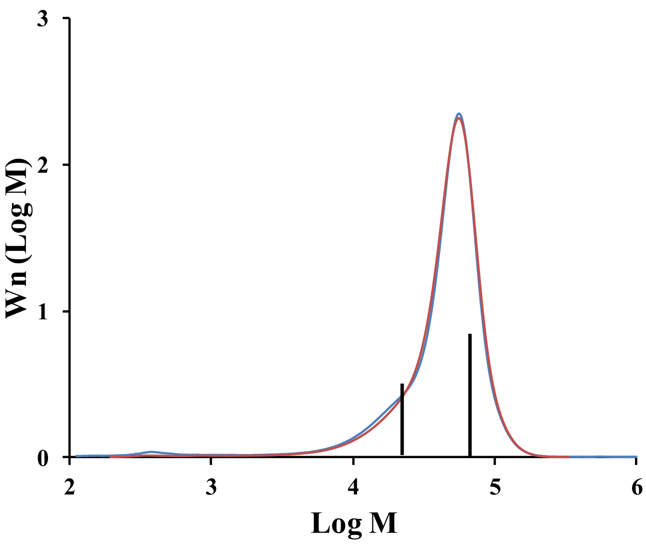

Figure 5. SEC curves of S-S/MA (one-step method) in THF/FA. S-S/MA block copolymer is shown with UV detection (270 nm) in blue and DRI detection in red. The curves are normalized such that each represents the total polymer weight. Resolved, discrete M_w values from the DECRA analysis of the material in THF-d_8/FA are shown as vertical lines with heights proportional to component concentrations (from).

weight. Furthermore, the molecular weight clearly has increased relative to the two-step synthesis. DECRA results model the data with two resolved components whose spectra are very similar. Qualitatively, a marginal compositional drift with molecular weight is evident as slightly more MA relative to S is seen in the higher molecular weight component.

A small but consistent solvent effect is seen with the S-S/MA polymers. The weighted average value for M_w calculated from the two resolved components in THF-d_8 (49000,) is comparable to SEC, but that calculated for the polymer in acetone-d_6 is lower (37000,). Therefore, the solvent quality of THF/FA must be better than acetone for this polymer.

A closer analysis of the blocked character of the S-S/MA polymers is obtained from the ^{13}C NMR data. The aromatic carbon of the styrene closest to the polymer backbone is particularly sensitive to sequencing. The shift ranges for the four different triads with the sensitive quaternary aromatic carbon at the center are shown. In this case, total S/MA composition is more accurately obtained from the much narrower ^{13}C resonances compared to the broad ^{1}H resonances. By comparing the carbonyl integral with the total aromatic integral, we can obtain the S/MA molar composition. For the two-step S-S/MA, polymer S/MA = 59/41. Based upon the aromatic integral values of the DECRA analysis ( ), ~18% of the total S is homopolymer. The amounts of S block (S<u>S</u>S) and S/MA block (MA<u>S</u>MA) may be tallied from the triad distribution found in the ^{13}C NMR spectrum ( ) where 75% of the total S is found in the S/MA alternating block. The final material from S-S/MA step 2 may be represented as S_{11} + S_4-*b*-(S_{44}-*alt*-MA_{42}) (molar

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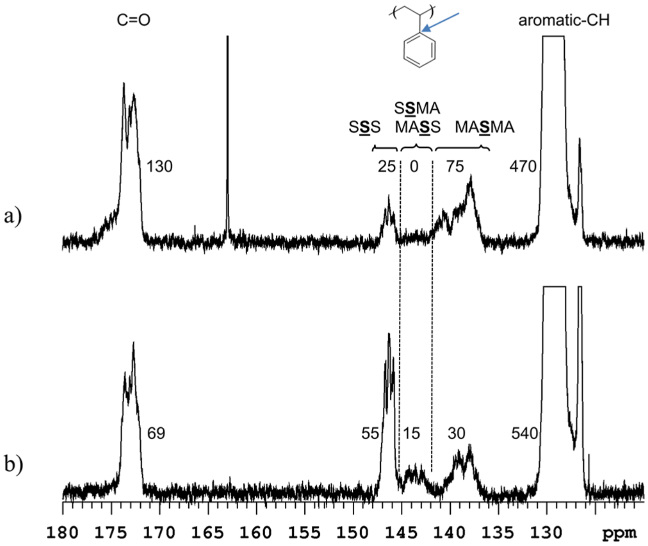

Figure 7. Quantitative [13]C NMR spectra showing the aromatic and carbonyl region of (a) S-S/MA (two-step method) and (b) S-S/MA (one-step method). The aromatic carbon attached to the backbone is sensitive to chain sequencing and has a chemical shift range in these materials from 136 to 148 ppm. Integral values are shown.

ratio). Again, we conclude that most of the added material in the second step formed S homopolymer with very little growth of the S block.

The S/MA molar composition for the one-step S-S/MA polymer is 76/24 based upon the carbonyl and aromatic integrations of the ^{13}C NMR spectrum ( ). The presence of "semi-alternating" triads, SSMA and MASS (in the region 141−145 ppm), for the one-step S-S/MA spectrum is indication of random structure. The SSS resonance (145−148 ppm) indicates that 55% of the total S is in the blocked form. The final material from S-S/MA single step may then be represented as S_{42}-*b*-S_{34}/MA_{24} (molar ratio). The fact that S and MA are unequal in the S/MA block indicates that the block is not strictly alternating. A more random S/MA structure must exist in which single MA monomers are dispersed within small blocks of S. Considering that 30% of the S in the S/MA block is in the alternating sequence, MASMA (136−141 ppm), the polymer may tentatively be represented as S_{42}-*b*-(S_{11}-*ran*-MA_1)-*b*-(S_{23}-*alt*-MA_{23}) (molar ratio). This result gives a random S/MA portion that is over 90% S and leaves us with the conclusion that this simple accounting approach is speculative. However, it is consistent with previous work. Ha found that small molar fractions of MA, lower than 0.05, in S/MA polymers show significant signal in the "semi-alternating" region of the spectrum. This high sensitivity to sequencing makes it difficult to pinpoint the exact structure. The complexity may thus be explained by a structure bridging the S and S-*alt*-MA blocks that is either a tapered or a random array of small blocks of S-*alt*-MA and S. There is virtually no way for NMR to distinguish the two blocks. A pure linear diblock S-*b*-(S-*alt*-MA) would show both little signal in the "semi-alternating" region of the ^{13}C NMR spectrum and similar DECRA-resolved components.

4. CONCLUSIONS

RAFT polymerizations with the aim of forming linear block polymers have been examined with three analytical methods: multidetection SEC, 1D NMR (^{1}H NMR and ^{13}C quantitative NMR), and PGSE NMR with DECRA analysis. These methods provide a comprehensive characterization including chain growth, overall chemical composition, chemical composition heterogeneity, and detection/quantification of homopolymer formation. The polymer samples provided with minimal purification were shown to be mixtures through DECRA analysis of ^{1}H PGSE NMR data. Resolved polymeric components based upon hydrodynamic size in solution identified both the presence of a homopolymer and compositional heterogeneity. Thus, the results provide a significantly different materials characterization compared with the simple 1D NMR analysis. M_w was determined for DECRA-resolved spectral components based upon the scaling (M_w vs R_h) of PEO in D_2O at 25 °C and compared directly to the M_w distributions obtained from SEC. The M_w comparison between SEC and NMR was not exact, which is attributed to the use of different polymer calibrations for each technique. Synthetic approaches involving a two-step procedure yielded homopolymer impurities. The one-step synthetic approach yielded no homopolymer but led to considerable chemical compositional heterogeneity identified by quantitative ^{13}C NMR. These methods are used routinely in our laboratory to expedite the synthetic effort and to guide further purification steps.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the at DOI: .

Details of the DECRA algorithm with further references; all data mentioned but not presented in the main article, including the DECRA-processed ^{1}H PGSE NMR data from , the ^{1}H NMR spectrum of the S-AM polymer, and UV spectra of both S and S-AM ( )

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The authors thank Joseph Lott for very helpful discussions.

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DOI:
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